

Alert: Anomaly detected in ASEP registry

Endpoint: ec2amaz-9oh9aqq

User: NT AUTHORITY\SYSTEM

OS: WindowsServer2022

Verdict: True positive

Alert story

[Maxi](#)

The screenshot displays an alert timeline with the following entries:

- Process tree:** [6820] powershell.exe -ExecutionPolicy Bypass -C "New-Item 'HKCU:\Software\Classes\ms-sett..."
- Alert timeline:**
 - Suspicious PowerShell command line** (Medium, Detected, New)
 - Suspicious process executed PowerShell co...** (Medium, Detected, New)
 - powershell.exe executed a script**
 - [6820] powershell.exe chan... .DEFAULT\Software\Classes\ms-settings\Shell\Open\comma...**
 - Anomaly detected in ASEP registry** (Medium, Detected, Resolved)
 - UAC bypass was detected** (Medium, Detected, New)
 - Defender detected and terminated active 'Behavior:Win32/UACBypassExp.T...** (Malware)
 - Suspicious 'UACBypassExp' behavior was bl...** (Low, Blocked, New)
 - [7740] fodhelper.exe**
- Additional related files:**
 - wusvc.exe** (Malware)
 - Anomaly detected in ASEP registry** (Medium, Detected, Resolved)

Command Line:

Some of command are:

```
powershell.exe -ExecutionPolicy Bypass -C "New-Item 'HKCU:\Software\Classes\ms-  
settings\Shell\Open\command' -Force | Out-Null; Set-ItemProperty 'HKCU:\Software\Classes\ms-  
settings\Shell\Open\command' '(Default)' \"cmd /c whoami /all > $env:TEMP\svc_ident.dat\"; New-  
ItemProperty 'HKCU:\Software\Classes\ms-settings\Shell\Open\command' DelegateExecute \" -Force  
| Out-Null; Start-Process fodhelper.exe -WindowState Hidden"
```

```
New-Item 'HKCU:\Software\Classes\ms-settings\Shell\Open\command' -Force | Out-Null; Set-  
ItemProperty 'HKCU:\Software\Classes\ms-settings\Shell\Open\command' '(Default)' "cmd /c  
whoami /all > $env:TEMP\svc_ident.dat"; New-ItemProperty 'HKCU:\Software\Classes\ms-  
settings\Shell\Open\command' DelegateExecute \" -Force | Out-Null; Start-Process fodhelper.exe -  
WindowState Hidden
```

Why was the alert triggered?

A process registered a suspicious command or file in ASEP registry key, where it will be run after a reboot.

An attacker may place a malicious piece of software in such a location to prevent losing access if a machine is turned off.

Investigation Summary

Microsoft Defender for Endpoint generated a high-severity alert for **Anomaly detected in ASEP registry** on **ec2amaz-9oh9aqq**. Investigation identified powershell.exe executing with - ExecutionPolicy Bypass to modify the registry key HKCU\Software\Classes\ms-settings\Shell\Open\command, create the DelegateExecute value, and launch fodhelper.exe. This sequence is consistent with the **Fodhelper UAC bypass** technique used to escalate privileges.

The activity originated from the suspicious executable **C:\Users\Public\wusvc.exe**, which communicated with **172.31.2.203:8888** and was detected by Microsoft Defender as **Trojan:Win64/SandCat!rfn**. Defender also detected **Behavior:Win32/UACBypassExp.T!gen** and successfully blocked and remediated the malicious behaviour. The alert is assessed as a **True Positive** indicating attempted **Privilege Escalation**, **Persistence**, and **Defense Evasion**.

MITRE ATT&CK:

- T1548.002 – Bypass User Account Control
- T1112 – Modify Registry
- T1547 – Boot or Logon Autostart Execution (ASEP)
- T1059.001 – PowerShell

Remediation

- Isolated the affected endpoint from the network.
- Removed C:\Users\Public\wusvc.exe
- Deleted the malicious registry keys (HKCU\Software\Classes\ms-settings\Shell\Open\command and DelegateExecute)
- Conducted a full antivirus scan of the endpoint.
- Investigated **172.31.2.203:8888** connections, file hashes, filenames (wusvc.exe), registry modifications, similar PowerShell command lines and isolated the affected hosts.
- Continued monitoring for any recurrence of UAC bypass or related malicious activity.